Q.1 (a)

Define the following terms. 1) Metadata 2) Schema 3) Data dictionary.

Q.1 (b)

Write down advantages of Database Management system.

Q.1 (c)

Explain Responsibilities of DBA.

OR Q.1 (c)

What is data abstraction? Explain three level ANSI SPARC architecture in detail.

Q.1 (a)

Explain three-level database architecture.

Q.1 (b)

Explain Total Participation and Partial Participation with example.

Q.1 (c)

Explain advantages of DBMS over file management systems.

OR Q.1 (c)

List out various data models. Explain any two in brief.

Q.2 (a)

Differentiate Schema vs Instance

Q.2 (b)

Explain Specialization with example.

Q.2 (c)

What is ER diagram? Explain different symbols used in E-R diagram with example.

OR Q.2 (a)

Differentiate DA vs DBA.

OR Q.2 (b)

Explain Generalization with example.

OR Q.2 (c)

What is attribute? Explain different types of attributes with example.

Q.2 (a)

Explain Mapping Cardinalities.

Q.2 (b)

Explain Outer Join operation in Relational Algebra.

Q.2 (c)

Explain concept of Specialization and Generalization with example.

OR Q.2 (a)

Explain different types of Keys in Relational Algebra.

OR Q.2 (b)

Explain types of attributes in ER-diagram with suitable example.

OR Q.2 (c)

Explain SELECT, PROJECT, UNION and SET-INTERSECTION operation with suitable example.

Q.3 (a)

Explain the GRANT and REVOKE statement in SQL.

Q.3 (b)

Explain following Character functions. 1) INITCAP 2) SUBSTR

Q.3 (c)

Consider following tables and write answers for the given queries.
stud_master (enroll_no, name, city, dept)

1. Display all the student detail who study in IT dept
2. Retrieve all information about name where name begin with ‘p’.
3. Insert new student to a table.
4. Add a new column gender to a table stud_master.
5. Count the number of rows for stud_master table.
6. Display all the student detail in descending order of enroll_no
7. Destroy table stud_master along with data.

OR Q.3 (a)

Explain equi join with example in SQL.

OR Q.3 (b)

Explain following Aggregate functions. 1) MAX 2) SUM

OR Q.3 (c)

Write SQL queries for the following table:
PRODUCT_Master (prod_no, prod_name, profit, quantity, sell_price, cost_price)

1. Create table PRODUCT_Master.
2. Insert one record in this table.
3. Find out product having profit greater than 20000.
4. Delete product having quantity less than 5.
5. Add 2% profit in product having sell price is greater than 5000.
6. Add new field total_price to PRODUCT_Master.
7. Find out product name having no duplicate data.

Q.3 (a)

Differentiate Primary Key and Foreign Key constraint.

Q.3 (b)

Explain DUAL table and SYSDATE with example.

Q.3 (c)

Write SQL queries to use various numeric functions (integer, absolute value, ceil, floor, sqrt, 12^6, mod, sign).

OR Q.3 (a)

Explain Unique and Check Constraint with suitable example.

OR Q.3 (b)

Explain structure of PL/SQL block.

OR Q.3 (c)

Consider the following table and solve queries:
BRANCH (branchid, branchname, address)
EMPLOYEE (empid, name, post, gender, birthdate, salary, branchid).
I) Create the BRANCH table with branchid as PRIMARY KEY

II) Create the EMPLOYEE table with empid as PRIMARY KEY and branchid as Foreign Key
III) Find employees working in Ahmedabad branch
IV) Find employees born in 1998
V) Find female employees with salary more than 5000
VI) Find the address of the branch where Ajay is working

Q.4 (a)

Explain fully functional dependency with example.

Q.4 (b)

Relational Algebra Expressions for Employee table queries.

Q.4 (c)

What are the criteria of 2NF? Normalize given table into 2NF.

OR Q.4 (a)

Explain 3NF with example.

OR Q.4 (b)

Relational Algebra questions for Students (Name, SPI, DOB, Enrollment No)

OR Q.4 (c)

What are the criteria of 1NF? Normalize given table into 1NF with two different techniques.

Q.4 (a)

Explain Referential Integrity with suitable example.

Q.4 (b)

Differentiate Partial and Full Functional Dependency.

Q.4 (c)

Explain 3rd Normal Form with example.

OR Q.4 (a)

Explain Importance of Normalization.

OR Q.4 (b)

Differentiate Prime Attributes and Non-Prime Attributes.

OR Q.4 (c)

Explain 2nd Normal Form with example.

Q.5 (a)

Explain ACID properties of transaction.

Q.5 (b)

Create STUDENT table with constraints:
(i) stu_name as not null
(ii) Branch must be IT/Computer/Electrical/Civil

Q.5 (c)

What is trigger? Write syntax & create trigger before inserting record.

OR Q.5 (a)

Explain problems of concurrency control in transaction.

OR Q.5 (b)

Create STUDENT table with constraints:
(i) stu_id as primary key
(ii) stu_id must start with 'S'

OR Q.5 (c)

What is Explicit cursor? Explain with example.

Q.5 (a)

Explain Transaction states with proper diagram.

Q.5 (b)

Explain any two DDL commands with suitable example.

Q.5 (c)

Explain ACID Properties in detail.

OR Q.5 (a)

What is two phase locking technique?

OR Q.5 (b)

Explain any two DML commands with suitable example.

OR Q.5 (c)

List problems of concurrency control and explain any two in detail.
